

BANK MARKETING CAMPAIGN ANALYSIS USING POWER BI

Power BI Data Analytics Project Documentation

1. PROJECT OVERVIEW AND OBJECTIVE

- The **bank marketing campaign analysis** project focuses on analysing bank marketing campaign data and transforming raw customer and campaign information into meaningful business insights using **power bi**.
- The project demonstrates an end-to-end data analytics workflow, including **data preparation, data cleaning, data transformation, data modelling, DAX calculations, interactive visualizations, and business insight generation**.
- The main objective of this project is to evaluate the effectiveness of bank marketing campaigns and understand the factors that influence customer subscription outcomes. The interactive power bi dashboard allows users to analyse campaign performance based on customer characteristics and campaign-related factors such as **call duration and campaign contact frequency**.
- The analysis helps identify patterns in customer responses and provides data-driven insights that can support better marketing strategies, customer targeting, and campaign decision-making.

2. DATA SOURCE

The project uses a **bank marketing dataset** containing customer-related and marketing campaign-related information.

The dataset includes information about customer demographics, financial characteristics, previous campaign interactions, current campaign contacts, call duration, and subscription outcomes.

Dataset Name: Bank Marketing

Source: UCI Machine Learning Repository

Domain: Banking / Marketing Analytics

Task: Classification

Dataset Type: Multivariate

Features: 16 Input Features

Instances: 45,211

3. BUSINESS PROBLEM

Bank marketing campaigns generate large amounts of customer and campaign interaction data. Analysing this data manually can make it difficult to identify the factors that influence customer subscription and campaign success.

The key business questions addressed in this project are:

- What is the overall customer subscription rate?
- How effective is the bank's marketing campaign?
- Does call duration have an impact on customer subscription?
- How does the number of campaign contacts affect subscription performance?
- Which customer segments show better campaign responses?
- What patterns can help improve future marketing campaigns?
- How can the bank improve customer targeting and campaign efficiency?

4. PROJECT OBJECTIVES

The main objectives of this project are:

- To clean and prepare the bank marketing dataset for analysis.
- To transform raw data into an analysis-ready format.
- To analyse customer subscription and campaign performance.
- To calculate important marketing KPIs using DAX.
- To analyse subscription rates based on call duration.
- To analyse subscription rates based on campaign contact frequency.
- To identify customer segments with stronger marketing responses.
- To develop an interactive power bi dashboard.
- To provide descriptive and diagnostic insights from the campaign data.
- To provide actionable recommendations for improving future marketing campaigns.
- To support data-driven decision-making using business intelligence techniques.

5. ATTRIBUTE / COLUMN DETAILS

Attribute name	Data type	Description
Age	Numeric	Age of the customer.
Job	Categorical	Occupation or job category of the customer.
Marital status	Categorical	Marital status of the customer.
Education	Categorical	Education level of the customer.
Default	Categorical	Indicates whether the customer has a credit default.
Balance	Numeric	Customer account balance.
Housing loan	Categorical	Indicates whether the customer has a housing loan.
Personal loan	Categorical	Indicates whether the customer has a personal loan.
Contact	Categorical	Communication method used to contact the customer.
Day / month	Date-related	Date or period information related to the campaign contact.
Call duration	Numeric	Duration of the marketing call.
Campaign	Numeric	Number of contacts made with the customer during the campaign.
Previous contacts	Numeric	Number of contacts made during previous marketing campaigns.
Subscription / target	Categorical	Indicates whether the customer subscribed to the promoted product or service.

6. TOOLS & TECHNOLOGIES

The following tools and technologies were used in this project:

Microsoft excel / power query

- Data inspection

- Data cleaning
- Data transformation
- Data filtering
- Data preparation

Power BI

- Data modelling
- Dax calculations
- KPI creation
- Interactive dashboard development
- Data visualization
- Business insight generation

Dax

Dax was used to create analytical measures and calculate important KPIs such as:

- Total customers
- Total subscribers
- Non-subscribers
- Subscription rate
- Average call duration
- Average campaign contacts
- Subscription rate by call duration
- Subscription rate by campaign contact frequency

7. DATA PRE-PROCESSING

The raw bank marketing data was prepared and transformed before creating the power BI dashboard.

Data cleaning and transformation

The following data preparation activities were performed:

- Reviewed the raw dataset and identified relevant customer, campaign, and subscription fields.
- Checked and standardized data types.

- Identified missing or inconsistent values where applicable.
- Removed duplicate records where required.
- Filtered and organized relevant records for analysis.
- Standardized categorical values for consistent analysis.
- Created derived analytical fields to improve dashboard usability.
- Created **call duration categories** to analyse subscription performance across different call-duration groups.
- Prepared the dataset for power BI data modelling and visualization.

Data transformation

The data was transformed into a suitable structure for analysis. Calculated fields and categories were created where necessary to support meaningful comparisons and interactive dashboard analysis.

8. DATA MODELLING AND DAX

The prepared dataset was loaded into power bi for data modelling and analysis.

The power bi model was designed to support analysis of:

- Customer characteristics
- Campaign performance
- Contact frequency
- Call duration
- Subscription outcomes

Dax measures were created to calculate important KPIs and performance indicators.

Key measures

The dashboard includes analytical metrics such as:

- Total customers
- Total subscribers
- Total non-subscribers
- Subscription rate (%)
- Average call duration
- Average campaign contacts

- Subscription rate by call duration category
- Subscription rate by campaign contact frequency

Subscription rate

The **subscription rate** measures the percentage of customers who subscribed after being contacted during the marketing campaign.

This KPI helps evaluate the overall effectiveness of the bank's marketing campaign.

Call duration category

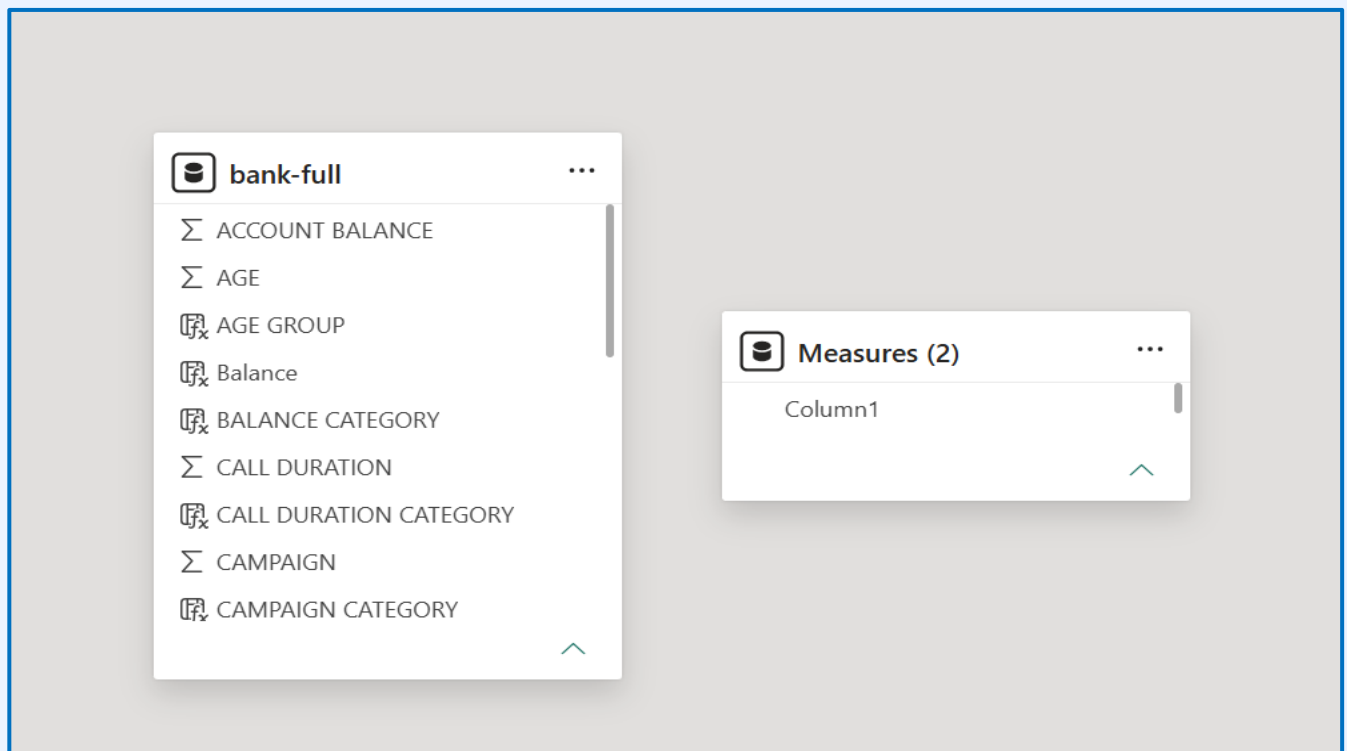
Call duration was converted into categories to make it easier to compare subscription performance across different call-duration groups.

This allows the business to understand whether different levels of customer interaction are associated with different subscription outcomes.

Data modelling

The data model and relationships were configured in power bi based on the structure of the available dataset.

Screenshot:



9. ANALYSIS AND VISUALIZATIONS

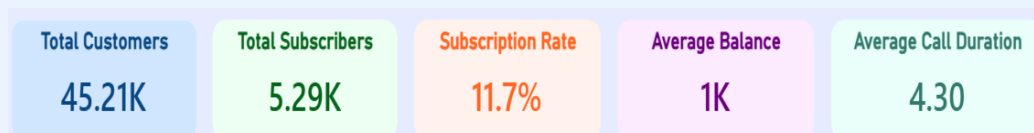
The power BI dashboard was designed as an interactive analytical report to help users understand customer subscription behaviour and marketing campaign performance.

Dashboard features

1. KPI cards

KPI cards display high-level campaign performance indicators such as:

- Total customers
- Total subscribers
- Subscription rate
- Average call duration
- Average Balance



These KPIs provide users with a quick overview of campaign performance.

2. Subscription rate analysis

This visual provides an overview of the percentage of customers who subscribed to the promoted product or service.

It helps stakeholders understand the overall success rate of the marketing campaign.

3. Subscription rate by call duration

This visual compares the subscription rate across different call-duration categories.

It helps identify whether customer response changes based on the duration of interaction between the bank representative and the customer.

This analysis can help the bank understand customer engagement and improve the effectiveness of marketing conversations.

4. Subscription rate by campaign contact

This visual analyses the relationship between the number of campaign contacts and customer subscription rate.

It helps determine whether repeated customer contact is associated with better subscription outcomes or whether excessive contact may have limited benefits.

5. Customer segmentation analysis

Customer attributes can be analysed using interactive visuals to identify segments with stronger or weaker campaign responses.

Potential segmentation dimensions include:

- Age
 - Job
 - Education
 - Marital status
 - Housing loan
 - Personal loan
 - Financial characteristics
-

6. Campaign performance analysis

Campaign-related attributes are analysed to identify patterns in customer response and subscription behaviour.

This helps stakeholders evaluate the effectiveness of different campaign approaches.

7. Slicers and filters

Interactive slicers and filters allow users to dynamically analyse the dashboard based on available customer and campaign dimensions.

This improves the usability of the dashboard and allows users to perform focused analysis.

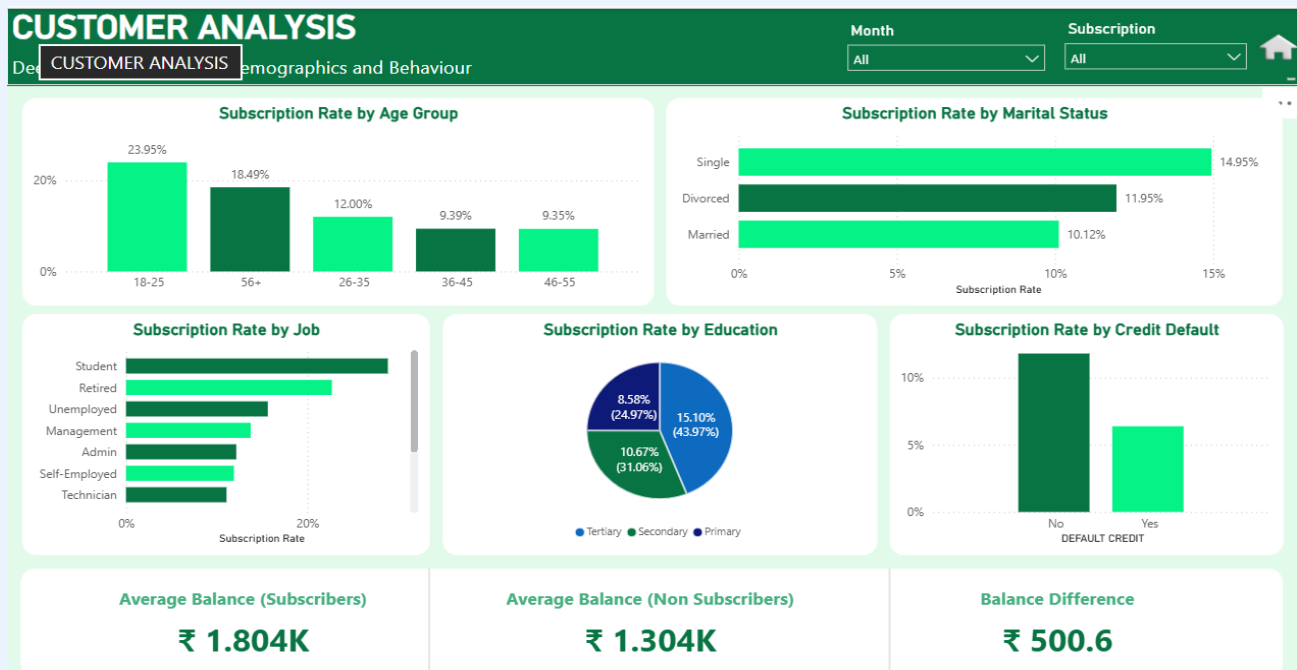
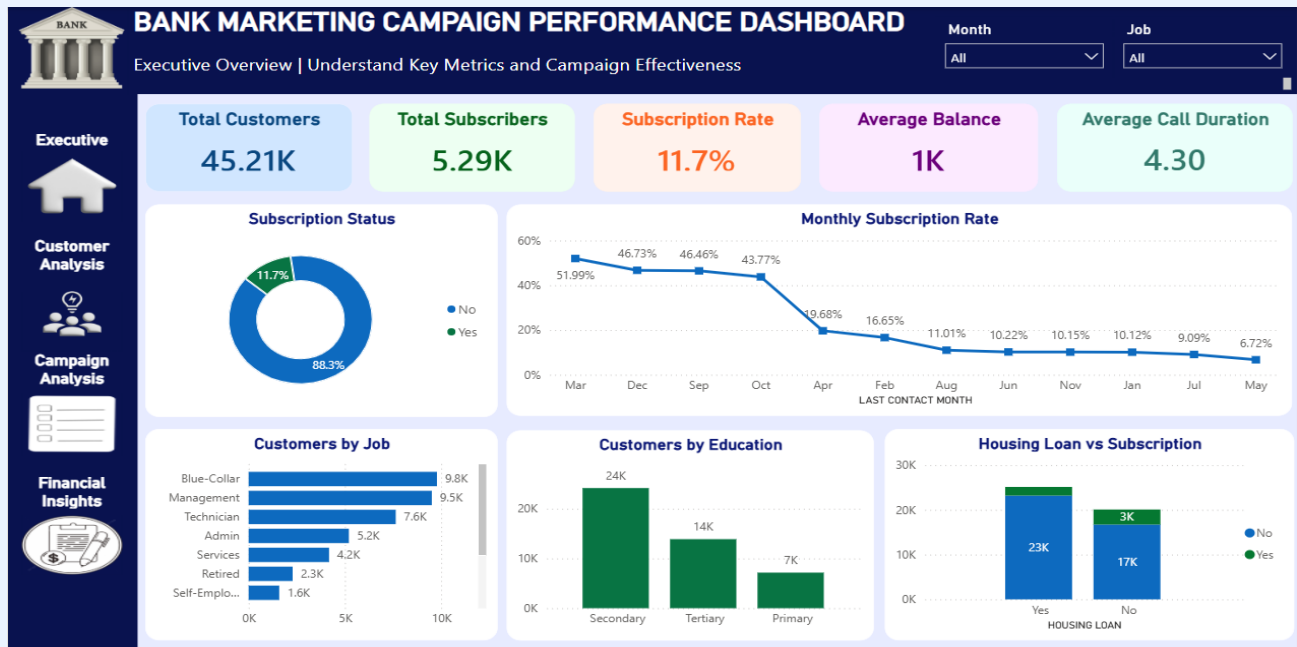
8. Interactive dashboard

The dashboard uses power BI's interactive capabilities, including:

- Slicers
- Filters

- Cross-filtering
- Interactive charts
- KPI cards
- Data labels
- Clear visual titles
- Dashboard navigation where applicable

Screenshot:



CAMPAIGN ANALYSIS

Understand Campaign Performance and Contact Effectiveness

CONTACT METHOD

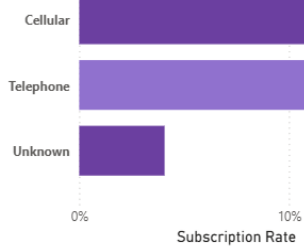
All

Month

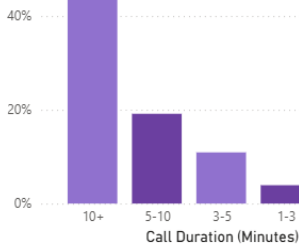
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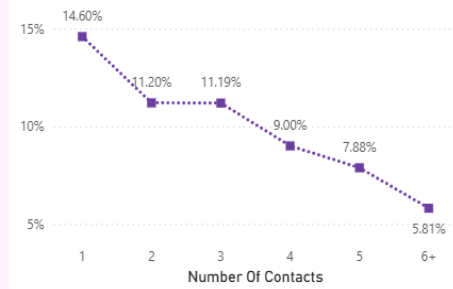
Subscription Rate by Contact Type



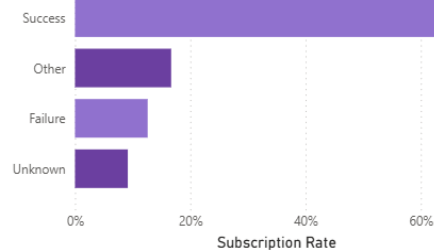
Subscription Rate by Call Duration



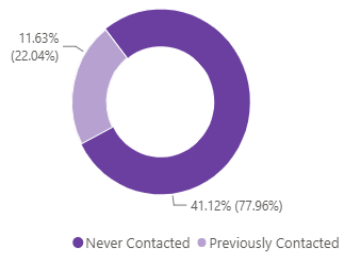
Subscription Rate by Campaign Contacts



Subscription Rate by Previous Outcome



Subscription Rate by Previous Contact Status



Average Days Since Previous Contact

40.20

500+ Days = Not Previously Contacted

Average Previous Contact

0.58

Per Customer

FINANCIAL INSIGHTS

Analyze Customer Balance and Financial Behaviour

Job

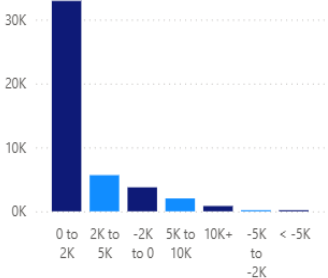
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Subscription

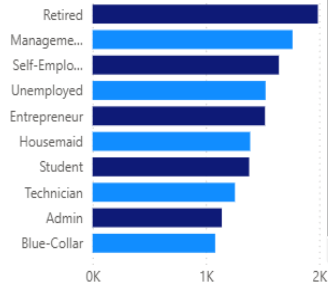
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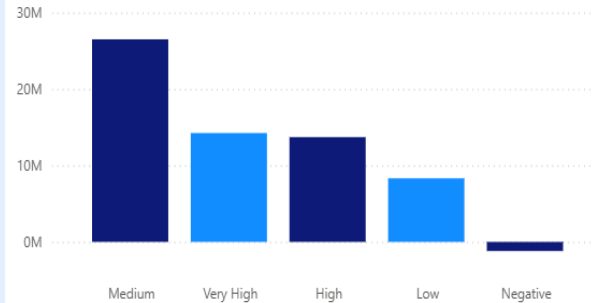
Balance Distribution



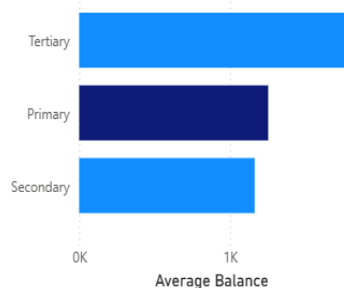
Average Balance by Job



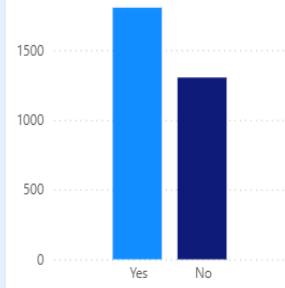
Total Balance by Balance Category



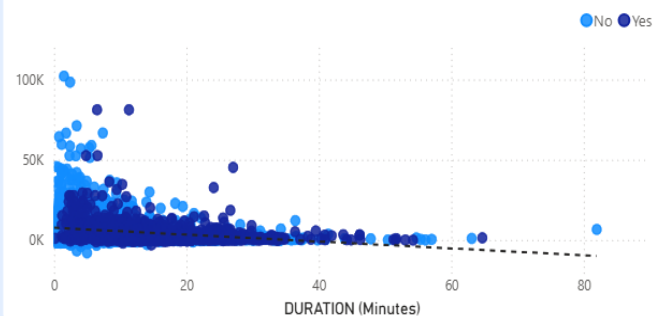
Average Balance by Education



Average Balance by Subscription Status



Balance vs Call Duration



10. INSIGHTS & CONCLUSIONS – SMART ANALYTICAL APPROACH

The dashboard follows a smart and analytical approach to convert raw data into actionable business insights.

A. Descriptive analysis

The descriptive analysis provides an overview of the campaign's performance.

Key areas analysed include:

- Total number of customers contacted.
- Total number of successful subscriptions.
- Overall subscription rate.
- Average call duration.
- Average campaign contact frequency.
- Customer and campaign segment performance.

The dashboard provides a consolidated view of the campaign and allows stakeholders to quickly understand the overall marketing performance.

B. Diagnostic analysis

The diagnostic analysis focuses on understanding the possible reasons behind differences in campaign performance.

Call duration analysis

The subscription rate by call duration helps identify whether customer response varies based on the length of the marketing conversation.

This analysis can help determine whether longer customer interactions are associated with stronger subscription outcomes.

Campaign contact analysis

The subscription rate by campaign contact frequency helps identify how repeated contacts influence customer response.

This can help the bank understand whether additional campaign contacts improve conversion or whether repeated outreach produces diminishing returns.

Customer segment analysis

Customer segmentation helps identify groups that demonstrate stronger or weaker responses to marketing campaigns.

This can support more focused customer targeting in future campaigns.

C. Predictive analysis

Historical campaign data can be used to identify customer segments and campaign characteristics associated with higher subscription rates.

Based on observed patterns, the bank can identify potential high-response customer groups and prioritize them for future marketing campaigns.

This provides a foundation for developing more advanced predictive models in future analytics projects.

D. Prescriptive analysis

Based on the dashboard analysis, the following recommendations can be considered:

- Focus marketing campaigns on customer segments with stronger historical subscription performance.
- Use call-duration analysis to improve the quality and effectiveness of customer conversations.
- Review campaign contact frequency to balance customer engagement and campaign efficiency.
- Reduce unnecessary repeated contacts where they do not contribute to improved subscription outcomes.
- Use customer segmentation to develop more targeted marketing campaigns.
- Continuously monitor subscription rates and campaign KPIs to measure campaign effectiveness.
- Use power BI dashboards for regular monitoring and data-driven marketing decisions.

11. KEY BUSINESS INSIGHTS

The bank marketing power BI dashboard provides the following key analytical perspectives:

Insight 1 – campaign effectiveness

The subscription rate provides a clear measure of how effectively the marketing campaign converts contacted customers into subscribers.

Insight 2 – call duration

Analysing subscription rate by call duration helps identify patterns between customer interaction time and subscription behaviour.

Insight 3 – campaign contact frequency

Analysing subscription rate by campaign contact count helps determine the potential impact of repeated customer outreach.

Insight 4 – customer segmentation

Customer-level attributes can help identify segments that are more responsive to marketing campaigns.

Insight 5 – targeted marketing

The insights generated from the dashboard can support more focused and efficient marketing strategies by prioritizing customer groups with stronger response patterns.

12. BUSINESS IMPACT

The bank marketing campaign analysis dashboard transforms raw marketing data into an interactive decision-support solution.

The dashboard enables business users to:

- Monitor campaign performance.
- Track subscription KPIs.
- Analyse customer response patterns.
- Compare call-duration performance.
- Evaluate campaign contact frequency.
- Identify high-response customer segments.
- Support targeted marketing decisions.
- Improve campaign planning and resource allocation.
- Make data-driven business decisions.

The project demonstrates how power bi can be used to convert complex marketing data into clear, actionable business insights.

13. SKILLS DEMONSTRATED

This project demonstrates practical skills relevant to **data analyst, power BI developer, business analyst, and BI analyst** roles.

Technical skills

- Microsoft power bi
- Power query
- Dax
- Data cleaning
- Data transformation
- Data modelling
- KPI development
- Data visualization
- Interactive dashboard development

Analytical skills

- Descriptive analytics
- Diagnostic analytics
- Customer analytics
- Marketing analytics
- Campaign performance analysis
- Segmentation analysis
- Business problem solving
- Data-driven decision making

Business skills

- Translating business questions into analytical requirements
- Identifying meaningful KPIs
- Communicating insights through dashboards
- Developing actionable recommendations
- Presenting complex data in a simple and understandable format

14. CONCLUSION

- The **bank marketing campaign analysis using power BI** project demonstrates an end-to-end data analytics workflow, starting from data preparation and transformation and progressing to data modelling, DAX calculations, interactive visualization, and business insight generation.
- The power BI dashboard provides an effective way to analyse customer subscription behaviour and campaign performance. By analysing factors such as **call duration, campaign contact frequency, and customer segments**, the project provides useful insights that can support better marketing strategies and data-driven decision-making.
- From a professional perspective, this project demonstrates the ability to work with real-world business data, develop interactive bi dashboards, create meaningful KPIs, analyse business performance, and communicate actionable insights to stakeholders.

15. PROJECT SUMMARY

BANK MARKETING CAMPAIGN ANALYSIS | POWER BI

Analysed bank marketing campaign data to identify factors influencing customer subscription outcomes. Performed data cleaning and transformation, developed DAX-based KPIs, and created an interactive power BI dashboard to evaluate subscription rates across call duration, campaign contact frequency, and customer segments. Generated descriptive and diagnostic insights and provided data-driven recommendations to support targeted marketing strategies and campaign optimization.